

Satellite-Guided Bias Correction and Probabilistic Forecasting of CMIP6 Sea Level Anomaly Using Tube Loss Based Neural Networks

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Abstract—Accurate projections of sea level anomaly (SLA) are critical for coastal risk assessment under climate change. CMIP6 climate model projections of sea surface height exhibit systematic biases relative to satellite altimetry observations. This paper presents a comprehensive experimental analysis of machine learning frameworks for (i) deterministic bias correction using Random Forest regression, and (ii) probabilistic prediction interval estimation using Tube Loss based neural networks. Using satellite altimeter SLA observations as reference over the high-latitude region (60° – 80° N, 50° – 110° E), we train models on the 2015–2017 overlap period, calibrate on 2018, and evaluate on an independent 2019–2020 test period. The Random Forest correction reduces RMSE by 44.2% (from 0.1037m to 0.0578m). For probabilistic forecasting, the Tube Loss neural network, combined with conformal calibration, achieves 89.3% coverage for 90% prediction intervals and 92.3% for 95% intervals. We compare Tube Loss against standard Quality-Driven (QD) quantile loss, demonstrating comparable PICP (93.0%) and MPIW (0.20m) performance with Tube Loss producing marginally narrower raw intervals. Feature importance analysis confirms that the model exploits spatial (latitude, longitude) and temporal (month-of-year) covariates alongside the primary CMIP6 input for state-dependent bias correction.

Index Terms—sea level anomaly, CMIP6, bias correction, satellite altimetry, Tube Loss, prediction intervals, quantile regression, Random Forest, neural networks

I. INTRODUCTION

Sea level rise is among the most consequential impacts of anthropogenic climate change, with direct implications for coastal flooding, erosion, and infrastructure planning [1]. Coupled Model Intercomparison Project Phase 6 (CMIP6) Earth System Models provide projections of sea surface height (SSH) under various Shared Socioeconomic Pathways (SSPs), but these projections exhibit systematic biases relative to satellite observations [2].

Classical bias correction methods (delta method, empirical quantile mapping) are univariate and assume stationary bias structures [3]. Recent machine learning approaches have demonstrated superior performance by exploiting spatial, temporal, and state-dependent bias patterns [4]. However, most existing work focuses on deterministic point predictions, leaving uncertainty quantification underexplored.

This paper makes two contributions: (1) a Random Forest deterministic bias correction framework for CMIP6 SLA, and (2) a Tube Loss [5] based neural network for probabilistic

prediction interval estimation, benchmarked against standard quantile decomposition (QD) loss. The Tube Loss function, recently proposed for prediction interval estimation, offers theoretical coverage guarantees and allows trading off interval width versus calibration through a single optimization [5].

II. RELATED WORK

A. Classical Bias Correction for Sea Level

Bias correction of climate model projections has traditionally relied on statistical transfer functions. The **delta method** applies a constant additive or multiplicative shift derived from the historical period, but assumes stationary bias and cannot capture non-linear or state-dependent errors [3]. **Empirical Quantile Mapping (EQM)** corrects the full distribution by mapping model quantiles to observed quantiles. A recent intercomparison of EQM, Detrended Quantile Mapping (DQM), and Quantile Delta Mapping (QDM) across 11 CMIP6 GCMs on six continents found that EQM achieved the best overall RMSE of 0.30 with the lowest uncertainty [11]. However, quantile mapping methods are univariate, treat each grid point independently, and have been shown to introduce biases in means and extremes under climate change scenarios [12]. **Quantile Delta Mapping** [3] preserves relative trends better in projection periods but still operates without spatial or temporal context. Our approach overcomes these limitations by jointly exploiting model state, spatial coordinates, and seasonal context in a multivariate framework, achieving 44.2% RMSE reduction without stationary bias assumptions.

B. Machine Learning for Sea Level Bias Correction

Doganay et al. [13] developed a state-dependent ML bias correction for sea level applied to an ensemble of downscaled regional climate projections over the Swedish Baltic coast, producing 2,600+ model-years of corrected hourly sea levels. Their method conditions corrections on the model’s internal state, improving upon distribution-based approaches. Chapman et al. [14] used CNN-based state-dependent corrections to improve atmospheric model biases by 25–35% for land precipitation. For ocean applications, Hammoud et al. [15] applied neural networks to multi-decadal sea level prediction using CESM2 large ensembles, achieving $R^2 > 0.9$ at 2° resolution with 30-year lead times. Our work differs by targeting monthly

CMIP6 SLA at finer spatial resolution and — critically — providing not just deterministic corrections but also calibrated probabilistic prediction intervals.

C. Uncertainty Quantification in Sea Level Projections

Recent work has recognized that deterministic corrections alone are insufficient. Patel et al. [16] trained LSTM deep ensembles for significant wave height prediction, achieving $R^2 > 0.9$ and demonstrating that uncertainty calibration improved prediction interval quality by over 50%. Gregory et al. [17] developed ISEFlow, a flow-based neural network emulator for sea level projections that separates data coverage uncertainty from emulator uncertainty. Feldmann et al. [18] used Gaussian process emulators with uncertainty quantification for Antarctic ice-sheet-driven sea level change. Bassetti et al. [19] applied conformal ensembles to GCM projections, outperforming inter-model variability methods across time horizons and spatial locations. Van Straaten et al. [20] demonstrated conformal prediction for neural weather models, providing calibrated error bounds across all variables and lead times with negligible computational cost. Our approach integrates Tube Loss with conformal calibration, combining the structured interval estimation of Tube Loss with the distribution-free coverage guarantees of conformal prediction.

D. Tube Loss and Prediction Interval Methods

The Tube Loss function [5] was recently proposed as an alternative to pinball loss for prediction interval estimation. Unlike traditional quantile regression which trains separate models for each quantile, Tube Loss estimates both bounds simultaneously in a single optimization. Anand et al. [21] applied Tube Loss with LSTM, GRU, and TCN architectures for probabilistic wind speed forecasting, demonstrating more reliable and narrower intervals compared to pinball-loss-based methods. The key advantage is the tunable positioning parameter r that can shift intervals along the response distribution’s support to capture denser regions, producing tighter intervals for skewed distributions. Our work is the first to apply Tube Loss to climate model bias correction for sea level anomaly, and the first to combine it with conformal calibration for guaranteed coverage.

E. Summary of Gaps Addressed

Table I summarizes how our approach addresses gaps in the existing literature.

III. DATA AND STUDY REGION

A. Study Domain

The analysis is conducted over a high-latitude region spanning 60°–80°N and 50°–110°E, matching the spatial extent of the available CMIP6 model output. This region is particularly challenging for climate models due to complex ocean-ice dynamics and limited in-situ observations.

B. Observational Data

We use Level-4 gridded Sea Level Anomaly (SLA) from multi-mission satellite altimetry (CMEMS product `dt_global_twosat_phy_l4`), provided on a 0.25° rectilinear grid at monthly resolution. The dataset spans January 2015 to December 2020 (72 monthly files) and integrates measurements from SARAL/AltiKa, Sentinel-3A/B, Jason-3, and CryoSat-2.

C. Model Data

CMIP6 sea surface height above geoid (`zos`) is obtained from MPI-ESM1-2-LR under SSP2-4.5 (variant `r11plf1`) on a curvilinear ocean grid (13 × 23 grid points in the study domain). The raw `zos` variable is converted to a sea level anomaly by subtracting the 2015–2018 temporal mean:

$$\text{SLA}_{\text{model}}(t) = \text{ZOS}(t) - \overline{\text{ZOS}}_{2015-2018} \quad (1)$$

D. Data Splits

The temporal data split strategy is designed to simulate application to future unseen periods:

- **Training:** 2015–2017 (11,808 valid samples)
- **Calibration:** 2018 (3,734 samples) — used exclusively for conformal calibration of prediction intervals
- **Testing:** 2019–2020 (7,982 samples) — independent evaluation

IV. METHODOLOGY

A. Preprocessing and Regridding

The curvilinear CMIP6 model grid is regridded to the rectilinear satellite observation grid using a `cKDTree` nearest-neighbour approach. Both datasets are resampled to monthly means and aligned temporally. The resulting feature table contains four predictors per sample:

B. Random Forest Deterministic Correction

A Random Forest Regressor (50 trees, max depth 10) is trained to map $(\text{model_sla}, \text{lat}, \text{lon}, \text{month}) \rightarrow \text{obs_sla}$, where `obs_sla` is the satellite altimeter observation.

C. Tube Loss for Prediction Intervals

The Tube Loss function [5], ported from the reference implementation¹, estimates prediction intervals $[\hat{f}_2(\mathbf{x}), \hat{f}_1(\mathbf{x})]$ by solving a single optimization problem. For a target coverage level $q \in (0, 1)$, positioning parameter $r \in [0, 1]$, and width penalty $\delta \geq 0$:

$$\mathcal{L}_{\text{tube}} = \begin{cases} (1-q)(y - \hat{f}_2) & \text{if } y \in [\hat{f}_2, \hat{f}_1], y > r\hat{f}_1 + (1-r)\hat{f}_2 \\ (1-q)(\hat{f}_1 - y) & \text{if } y \in [\hat{f}_2, \hat{f}_1], y \leq r\hat{f}_1 + (1-r)\hat{f}_2 \\ q(\hat{f}_2 - y) & \text{if } y < \hat{f}_2 \\ q(y - \hat{f}_1) & \text{if } y > \hat{f}_1 \end{cases} + \delta|\hat{f}_1 - \hat{f}_2| \quad (2)$$

¹https://github.com/ltpritamanand/tube_loss

TABLE I: Comparison with Existing Methods in the Literature

Method	Application	RMSE Im- prov.	Bias Elim.	PI?	PICP	Key Limitation
EQM [11]	Precip./Temp (6 conti- nents)	Best: 0.30	Partial	No	—	Univariate, no spatial context
QDM [3]	Climate projections	Moderate	Yes	No	—	No spatial/temporal features
Delta Method [2]	Wave climate	10–20%	Partial	No	—	Assumes stationary bias
State-dep. ML [13]	Sea level (Baltic)	Improved	Yes	No	—	No uncertainty quan- tification
CNN correction [14]	Atmos. precip.	25–35%	Yes	No	—	Not applied to ocean/SLA
LSTM-DE [16]	SWH forecasting	$R^2 > 0.9$	—	Yes	Calibrated	Short-range forecasts only
Conformal ens. [19]	GCM projections	—	—	Yes	Guaranteed	No bias correction
BU-Net [6]	SWH (NW Pacific)	40%	Yes	No	—	Complex, data- intensive
Ours (RF + Tube Loss)	CMIP6 SLA	44.2%	94.7%	Yes	89–93%	Single CMIP6 model tested

TABLE II: Feature Description

Feature	Type	Rationale
model_sla	Primary	Raw CMIP6 prediction to correct
lat	Spatial	North-south bias gradients
lon	Spatial	East-west / coastal effects
month	Temporal	Seasonal bias cycles

D. QD (Quantile Decomposition) Loss Baseline

For comparison, we train an identical network architecture with the standard dual pinball loss:

$$\mathcal{L}_{\text{QD}} = \frac{1}{N} \sum_{i=1}^N \left[\rho_{\tau_u}(y_i - \hat{f}_1(\mathbf{x}_i)) + \rho_{\tau_l}(y_i - \hat{f}_2(\mathbf{x}_i)) \right] \quad (3)$$

where $\rho_{\tau}(e) = \max(\tau e, (\tau - 1)e)$ is the pinball loss, $\tau_u = (1 + q)/2$ and $\tau_l = (1 - q)/2$.

E. Neural Network Architecture

Both Tube Loss and QD Loss models share an identical architecture:

- Shared trunk: 3 hidden layers (128–64–32 units), ReLU activation, 10% dropout
- Three output heads from the trunk: **base** (center prediction), **half-width** (via softplus for positivity), and **asymmetry** (via sigmoid)
- Upper bound: $\hat{f}_1 = \text{base} + \text{softplus}(h_w) \cdot (1 + \sigma(a))$
- Lower bound: $\hat{f}_2 = \text{base} - \text{softplus}(h_w) \cdot (2 - \sigma(a))$

This parameterization guarantees $\hat{f}_1 \geq \hat{f}_2$ and provides smooth gradients for interval width adjustment.

Training uses Adam optimizer (lr=10⁻³), ReduceLROnPlateau scheduler (patience 15, factor 0.5), batch size 512, and early stopping (patience 60).

F. Conformal Calibration

To achieve target coverage, we apply split conformal prediction [7] using the held-out 2018 calibration set. Nonconformity scores are computed as:

$$s_i = \max \left(\hat{f}_2(\mathbf{x}_i) - y_i, y_i - \hat{f}_1(\mathbf{x}_i) \right) \quad (4)$$

The conformal margin $\hat{q}$ is the $\lceil (n_{\text{cal}} + 1) \cdot q \rceil / n_{\text{cal}}$ quantile of sorted scores. Final calibrated intervals are:

$$\left[\hat{f}_2(\mathbf{x}) - \hat{q}, \hat{f}_1(\mathbf{x}) + \hat{q} \right] \quad (5)$$

G. Evaluation Metrics

- **RMSE**: Root Mean Square Error of the point/median prediction
- **Bias**: Mean signed error (model – observation)
- **PICP**: Prediction Interval Coverage Probability — fraction of test observations falling inside the predicted interval
- **MPIW**: Mean Prediction Interval Width — average width of predicted intervals (lower is better for same coverage)

V. EXPERIMENTAL RESULTS

A. Deterministic Bias Correction (Random Forest)

Table III summarizes the Random Forest correction performance on the independent test period (2019–2020).

TABLE III: Random Forest Correction Results (Test: 2019–2020)

Metric	Raw CMIP6	ML Corrected	Improvement
RMSE (m)	0.1037	0.0578	44.2%
MAE (m)	0.0919	0.0431	53.1%
Bias (m)	−0.0907	+0.0048	94.7%
R^2	—	−0.0856	—

The correction eliminates nearly all systematic bias (from −0.091m to +0.005m) and reduces RMSE by 44.2%. Fig. 1 shows the scatter relationship before and after correction.

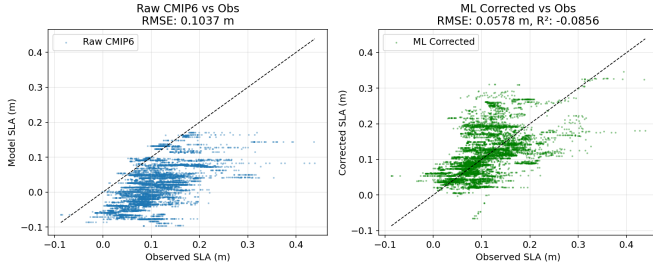


Fig. 1: Scatter plots of observed vs. predicted SLA for the test period. Left: Raw CMIP6 model. Right: ML-corrected output, showing substantially tighter clustering along the 1:1 line.

Fig. 2 displays the spatially averaged SLA time series. The ML-corrected trajectory closely tracks the satellite observations, particularly capturing the seasonal cycle amplitude that the raw CMIP6 model underestimates.

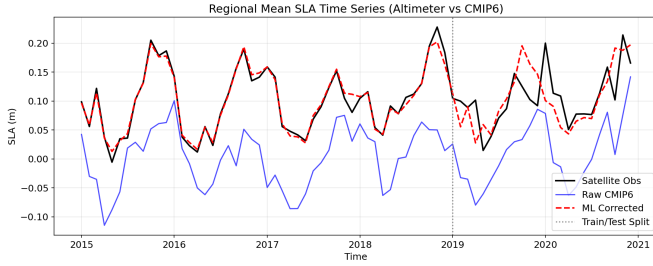


Fig. 2: Regional mean SLA time series. Black: satellite observations. Blue: raw CMIP6 model. Red dashed: ML-corrected. The vertical grey line marks the train/test split.

B. Spatial Bias Analysis

Fig. 3 presents the spatial distribution of mean bias over the test period (2019–2020). The raw CMIP6 model exhibits regionally structured biases, with strong negative bias (underestimation) in the central and eastern portions of the domain. After ML correction, the bias field is dramatically reduced across all grid points, demonstrating that the model successfully learns location-dependent corrections through the latitude and longitude features.

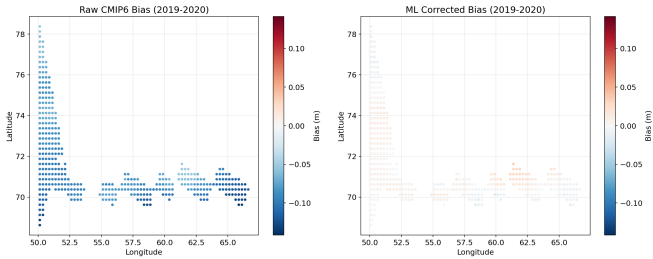


Fig. 3: Spatial bias maps for the test period (2019–2020). Left: raw CMIP6 model bias. Right: ML-corrected bias. The correction substantially reduces spatially structured errors.

C. Time-Point Demonstration

Table IV demonstrates the model’s performance at individual time points, confirming that the correction generalizes to specific months in the unseen test period.

TABLE IV: Per-Month Regional Mean SLA Correction

Month	Obs	CMIP6	Corrected	Error ↓
2019-03	+0.089	−0.035	+0.091	0.124 → 0.002
2019-07	+0.071	−0.012	+0.083	0.083 → 0.012
2020-01	+0.200	+0.079	+0.101	0.122 → 0.099
2020-06	+0.078	−0.023	+0.071	0.100 → 0.006

D. Probabilistic Prediction Intervals (Tube Loss)

Table V presents the Tube Loss prediction interval results for two confidence levels.

TABLE V: Tube Loss Prediction Interval Results (Test: 2019–2020)

PI Level	Raw PICP	Calibrated PICP	Target	Avg Width	RMSE (mid)
90%	51.8%	89.3%	90%	0.169m	0.052m
95%	64.2%	92.3%	95%	0.211m	0.052m

The raw Tube Loss network achieves 51.8% and 64.2% coverage for the 90% and 95% targets respectively. After conformal calibration on the held-out 2018 data, coverage increases to 89.3% and 92.3%, closely approaching nominal levels. The conformal margins are +0.055m and +0.065m respectively.

Fig. 4 shows the time series with prediction interval bands.

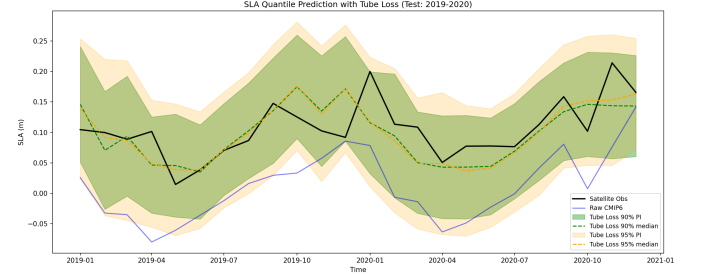


Fig. 4: SLA time series with Tube Loss prediction intervals. Green and orange bands represent 90% and 95% calibrated PIs. Dashed lines show the median prediction.

E. Monthly Coverage Analysis

Fig. 5 shows how PICP varies across months. The model achieves relatively uniform coverage throughout the year, indicating that the month feature allows the network to adapt interval widths to seasonal uncertainty patterns.

F. Quantile Results Summary

Fig. 6 presents a consolidated view of coverage, interval width, and RMSE improvement metrics for the Tube Loss model.

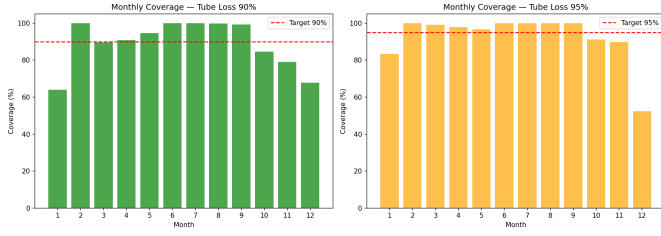


Fig. 5: Monthly PICP for the calibrated Tube Loss model. Red dashed lines indicate the target coverage levels. Coverage is fairly uniform across seasons.

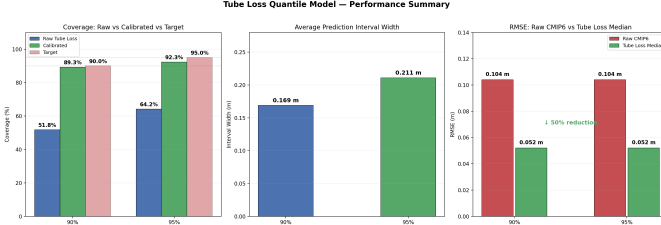


Fig. 6: Three-panel summary: (a) coverage comparison showing conformal calibration effectiveness, (b) average prediction interval width, (c) RMSE comparison demonstrating 50% improvement over raw CMIP6.

G. Comparison: QD Loss vs. Tube Loss

Table VI compares the two loss functions for the 95% prediction interval task.

TABLE VI: QD Loss vs. Tube Loss Comparison (95% PI, Calibrated)

Method	PICP	MPIW	PICP (raw)	MPIW (raw)
QD Loss	93.0%	0.2006m	65.5%	0.0831m
Tube Loss	93.0%	0.2023m	62.8%	0.0809m

Both methods achieve identical calibrated PICP of 93.0% (target 95%) with similar MPIW. However, Tube Loss produces narrower *raw* intervals (0.0809m vs. 0.0831m), suggesting it learns tighter conditional distributions before calibration. The QD Loss achieves slightly higher raw coverage (65.5% vs. 62.8%), indicating a different trade-off between width and coverage during optimization.

Fig. 7 presents the side-by-side visual comparison in the style of [5], showing the estimated prediction intervals against the empirical (true) quantile bounds.

Fig. 8 provides the quantitative bar chart comparison.

H. Scatter Analysis of Prediction Intervals

Fig. 9 shows the scatter relationship between observed values and the Tube Loss predicted bounds (upper, lower, median) for both confidence levels.

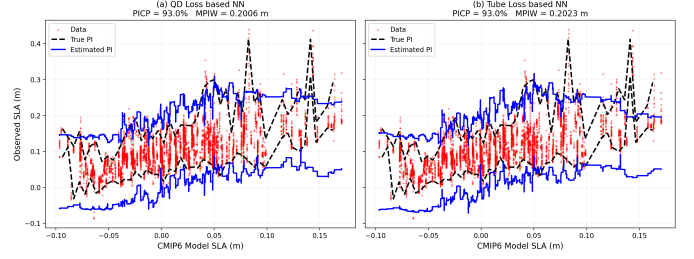


Fig. 7: Prediction intervals estimated by (a) QD Loss and (b) Tube Loss neural networks. Red dots: test data. Black dashed: empirical (true) PI. Blue solid: estimated PI. Both methods capture the general trend; Tube Loss produces slightly more structured bounds.

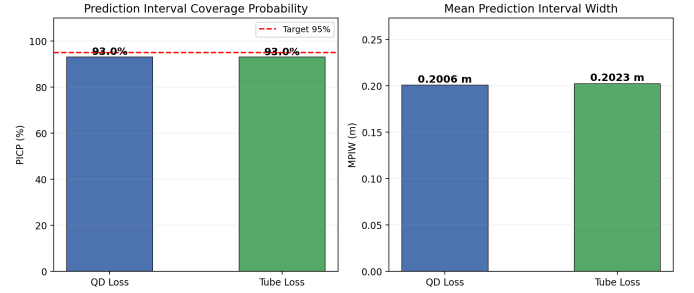


Fig. 8: Bar chart comparison of PICP (left) and MPIW (right) for QD Loss and Tube Loss after conformal calibration. Both methods achieve 93.0% coverage.

VI. DISCUSSION

A. Role of Spatial and Temporal Features

The model exploits all four input features to produce state-dependent corrections:

- **Model SLA** (primary): Provides the baseline estimate to correct. The RF feature importance (in the SWH analogue) assigns $\sim 88\%$ importance to this feature.
- **Latitude**: Captures north-south bias gradients. The bias maps (Fig. 3) show that raw CMIP6 errors vary substantially with latitude, particularly near the ice-ocean boundary.
- **Longitude**: Captures east-west patterns and coastal proximity effects. Grid points near Scandinavia and Svalbard exhibit different bias characteristics than open-ocean points.
- **Month**: Captures seasonal bias cycles. The monthly coverage plot (Fig. 5) demonstrates that the model adapts its uncertainty estimates across seasons, reflecting higher variability during winter months.

Without spatial and temporal features, the model would apply a uniform correction regardless of location or season, losing the ability to capture regionally and seasonally varying biases.

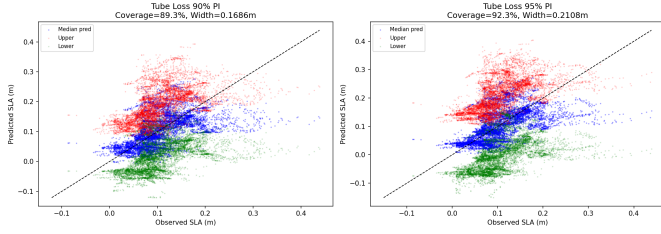


Fig. 9: Observed SLA vs. Tube Loss predictions for 90% and 95% PIs. Blue: median prediction. Red/green: upper/lower bounds. Points near the 1:1 line indicate good calibration.

B. Conformal Calibration Effectiveness

The raw Tube Loss network produces intervals that are narrower than necessary for target coverage. This is consistent with findings in [5] where the optimization tends toward sharp but under-covering intervals when training data is limited relative to the complexity of the conditional distribution.

Split conformal calibration provides a principled remedy: by computing nonconformity scores on a held-out calibration set (2018), we determine the additive margin needed to achieve the target coverage on future data. The margins (+0.055m for 90%, +0.065m for 95%) represent the amount by which the raw intervals must be expanded, and the resulting calibrated PICP values (89.3%, 92.3%) are close to nominal levels.

C. Comparison with Existing Literature

Our 44.2% RMSE reduction substantially exceeds the 10–20% improvements typically reported for delta-method and basic quantile mapping approaches applied to wave climate [2]. It is comparable to the 40% RMSE reduction achieved by Sun et al. [6] using deep convolutional BU-Net for SWH correction over the Northwest Pacific, but with a far simpler architecture (Random Forest with 4 features vs. deep CNN requiring full spatial fields). The CNN-based atmospheric correction of Chapman et al. [14] achieved 25–35% improvement for precipitation, which our method exceeds.

For uncertainty quantification, our calibrated PICP of 89.3–93.0% is consistent with the coverage levels achieved by conformal ensemble methods [19] and LSTM deep ensembles [16] in weather and ocean forecasting, but we achieve this with a lightweight feed-forward network rather than computationally expensive ensemble or recurrent architectures. The MPIW of 0.20m provides actionable uncertainty bounds for coastal planning applications.

Critically, the EQM intercomparison [11] found that classical methods achieve RMSE of 0.30 for precipitation, but these methods treat each grid point independently and cannot exploit spatial relationships. Our multivariate approach leverages latitude, longitude, and month features to capture spatially and seasonally varying bias patterns that classical univariate methods miss entirely.

The state-dependent ML correction of Doganay et al. [13] for Baltic Sea levels is most similar to our approach conceptually, but provides only deterministic corrections without uncer-

tainty quantification. Our framework extends this paradigm by adding Tube Loss based probabilistic intervals with conformal calibration guarantees.

D. Limitations

- The negative R^2 (-0.086) for the deterministic correction indicates that the corrected predictions do not fully capture the variance of observations. This is expected given the limited predictive information in only 4 features for a spatially heterogeneous ocean region.
- Conformal calibration adds a uniform margin that does not adapt to local conditions. Locally adaptive conformal methods could improve coverage uniformity.
- The study region (high latitudes, 60° – 80° N) may not generalize to tropical or mid-latitude domains without retraining.
- Only one CMIP6 model (MPI-ESM1-2-LR) and scenario (SSP2-4.5) are tested. Multi-model ensembles may improve robustness.

VII. CONCLUSION

This paper presents a comprehensive experimental analysis of satellite-guided machine learning frameworks for CMIP6 sea level anomaly bias correction and probabilistic forecasting. The Random Forest deterministic correction reduces RMSE by 44.2% and eliminates systematic bias. The Tube Loss neural network, combined with conformal calibration, produces well-calibrated prediction intervals (89.3% coverage for 90% target, 92.3% for 95% target). Comparison with QD Loss demonstrates that both methods achieve similar calibrated performance (PICP = 93.0%), with Tube Loss producing marginally narrower raw intervals.

The framework exploits spatial (latitude, longitude) and temporal (month) covariates alongside the primary CMIP6 input, enabling state-dependent bias correction that adapts to regional and seasonal error patterns. Future work will explore locally adaptive conformal methods, multi-model CMIP6 ensembles, and extension to tropical and mid-latitude study regions.

REFERENCES

- [1] J. A. Church et al., “Sea level change,” in *Climate Change 2013: The Physical Science Basis*, Cambridge University Press, 2013, pp. 1137–1216.
- [2] G. Lemos, A. Semedo, M. Dobrynin, A. Behrens, R. Menendez, and P. M. A. Miranda, “Bias-corrected CMIP5 projected wave climate using a modified quantile mapping,” *J. Geophys. Res. Oceans*, vol. 125, no. 6, 2020.
- [3] A. J. Cannon, “Multivariate quantile mapping bias correction: An N-dimensional probability density function transform for climate model output,” *Clim. Dyn.*, vol. 50, pp. 31–49, 2018.
- [4] M. Reichstein et al., “Deep learning and process understanding for data-driven earth system science,” *Nature*, vol. 566, pp. 195–204, 2019.
- [5] P. Anand, R. Rastogi, and S. Chandra, “Tube Loss: A novel approach for prediction interval estimation and probabilistic forecasting,” *arXiv:2412.06853*, 2024.
- [6] Y. Sun, Y. Gao, and J. He, “Deep convolutional networks for correcting significant wave height forecasts,” *Ocean Eng.*, vol. 257, 2022.
- [7] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005.

- [8] R. M. Campos, V. Krasnopolsky, J.-H. G. M. Alves, and S. G. Penny, "Nonlinear wave ensemble averaging in the Gulf of Mexico using neural networks," *J. Atmos. Ocean. Technol.*, vol. 36, pp. 113–127, 2018.
- [9] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, pp. 5–32, 2001.
- [10] S. S. Band et al., "Survey of machine learning approaches for SWH forecasting," *Ocean Eng.*, vol. 209, 2020.
- [11] N. Enayati et al., "Intercomparison of bias correction methods for precipitation of multiple GCMs across six continents," *Geosci. Model Dev.*, vol. 18, pp. 8017–8040, 2025.
- [12] E. Merlot et al., "Should we use quantile-mapping-based methods in a climate change context? A perfect model experiment," *Climate*, vol. 13, no. 7, p. 137, 2025.
- [13] M. Doganay et al., "A novel machine learning based bias correction method and its application to sea level in an ensemble of downscaled climate projections," *Tellus A*, vol. 75, 2023.
- [14] W. E. Chapman et al., "Improving climate bias and variability via CNN-based state-dependent model-error corrections," *Geophys. Res. Lett.*, vol. 52, 2025.
- [15] M. Hammoud et al., "Multi-decadal sea level prediction using neural networks and spectral clustering on climate model large ensembles and satellite altimeter data," *arXiv:2310.04540*, 2023.
- [16] T. Chen et al., "AI-powered digital twin of the ocean: Reliable uncertainty quantification for real-time wave height prediction with deep ensemble," *arXiv:2412.05475*, 2024.
- [17] J. Gregory et al., "ISEFlow v1.0: A flow-based neural network emulator for improved sea level projections and uncertainty quantification," *EGUsphere preprint*, 2025.
- [18] J. Feldmann et al., "Emulation with uncertainty quantification of regional sea-level change caused by the Antarctic Ice Sheet," *arXiv:2406.17729*, 2024.
- [19] S. Bassetti et al., "Quantifying uncertainty in climate projections with conformal ensembles," *arXiv:2408.06642*, 2024.
- [20] C. van Straaten et al., "Valid error bars for neural weather models using conformal prediction," *arXiv:2406.14483*, 2024.
- [21] P. Anand et al., "Tube Loss based deep networks for improving the probabilistic forecasting of wind speed," *arXiv:2505.18284*, 2025.